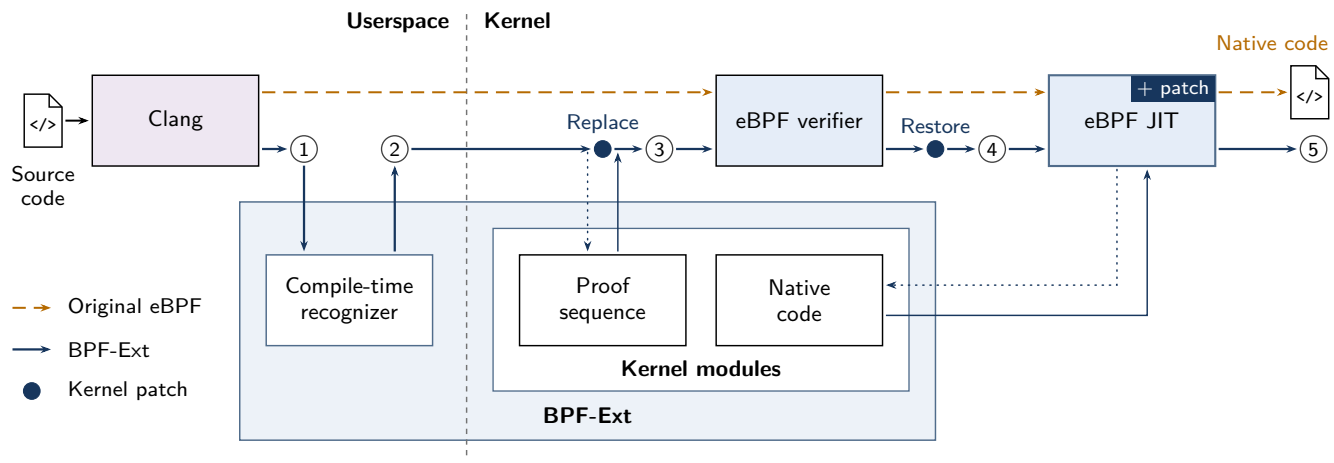


① Input eBPF bytecode

```
...
```

Rotate idiom

```
r1 = r7
r7 <<= 16
r1 >>= 48
r7 |= r1
```

```
r0 = r7
...
```

② Rewritten eBPF + KINSN

```
...
```

KINSN rotate

```
Sidecar
IMM, r7, r7, 16
```

```
Call
call d_rot
```

```
r0 = r7
...
```

③ Checked eBPF bytecode

```
...
```

Proof sequence

```
if r7 s<0 goto +2
r7 <<= 1
goto +2
r7 <<= 1
r7 |= 1
...  (16 steps)
```

```
r0 = r7
...
```

④ Restored eBPF + KINSN

```
...
```

KINSN rotate

```
Sidecar
IMM, r7, r7, 16
```

```
Call
call d_rot
```

```
r0 = r7
...
```

⑤ x86-64 Native code

```
...
```

Native code

```
rol r13, 16
```

```
mov rax, r13
...
```